

GARVIT GUPTA

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EDUCATION

B.Sc. (Hons.) Mathematics with Research, DDU College, Delhi University

Jul 2022 – Jun 2026

CGPA: **8.0 / 10** · Relevant coursework: Statistical Modeling, Linear Algebra, Optimization Theory, Probability & Stochastic Processes.

EXPERIENCE

ML & Mathematics Researcher — Final Year Thesis — DDU College, Delhi University

Aug 2025 – Jun 2026

- Designed a bias-mitigation algorithm combining statistical optimization with fairness-aware loss functions, reducing demographic bias up to 38% across 3 fairness metric families while holding accuracy within 1.2% of baseline.
- Built an EU AI Act-aligned Responsible AI platform (IBM AIF360, PyTorch, FastAPI, DagsHub) with a 5-phase automated bias-audit pipeline, cutting audit time from days to under 10 minutes across 20+ MLflow-tracked runs.

Machine Learning Intern — FastHire Manpower Solutions

Feb 2024 – Mar 2024

- Built an income classifier (Random Forest, SMOTE, GridSearchCV) on imbalanced socioeconomic data, reaching 91% accuracy and a 14% F1 improvement over the client's existing rule-based system.
- Shipped the model through a DagsHub CI/CD pipeline with automated experiment tracking, cutting the release cycle from 2 weeks to same-day.

PROJECTS

SCOUT — Supply Chain Disruption Early-Warning Agent [Live Demo ↗](#) · [GitHub ↗](#)

LangGraph · Gemini · XGBoost · MLflow/DagsHub · DVC · FastAPI · MySQL (Aiven) · HTML/CSS/JS

- To flag shipment delays before they happen, built a 3-agent LangGraph pipeline (Monitor → Risk → Recommend) scoring disruption risk and delay with two XGBoost models (ROC-AUC 0.76) trained on 180K+ real shipments.
- Turned scores into action: Gemini writes recommendations on a live FastAPI service and custom dashboard, with MLflow + DVC making every run reproducible.

Claim-Sentinel — Insurance Fraud Detection & Auto-Adjudication Agent [Live Demo ↗](#) · [GitHub ↗](#)

LangGraph · LightGBM · SHAP · Groq LLM · FastAPI · MySQL (Aiven) · HTML/CSS/JS

- To speed up claims triage without black-box decisions, built a 3-agent LangGraph pipeline (Intake → FraudRisk → Adjudication) on a class-balanced LightGBM model (ROC-AUC 0.83) with SHAP explanations for every prediction.
- Auto-approves low-risk claims and escalates high-risk ones with LLM-written, audit-logged adjuster notes; deployed on Render with cloud MySQL.

CropGuard — Crop Disease & Outbreak Detection App [GitHub ↗](#)

TensorFlow · MobileNetV2 · FastAPI · React · Groq · Gemini · Open-Meteo · Folium · Render

- To catch crop disease before it spreads, built a MobileNetV2 leaf classifier (PlantVillage) on FastAPI returning diagnosis, live weather risk, and Groq/Gemini treatment advice in English and Marathi, plus voice output.
- Made it reliable: 35% confidence gate rejects bad photos, cached model and weather with fallback providers (Groq → Gemini, Open-Meteo → wttr.in); flags outbreaks when 3+ farmers report the same disease in one district within 6 hrs.

Reverse-Engineering the Negation Circuit in GPT-2 (Mechanistic Interpretability) [GitHub ↗](#)

PyTorch · Activation Patching · Direct Logit Attribution · Causal Ablation

- To find what makes GPT-2 flip sentiment under negation, localized the responsible attention head using activation patching and direct logit attribution — two independent causal methods — across 960 minimal-pair prompts.
- Ablations proved the circuit is distributed and redundant: removing the top 10 heads cut 37% of the effect versus ~1% for the strongest single head.

AURUM — Gold Price Direction Intelligence [GitHub ↗](#)

FastAPI · Scikit-learn · MLflow · DagsHub Model Registry · Walk-Forward CV · Render.com · HTML/CSS/JS + Chart.js

- To forecast gold direction without look-ahead leakage, trained an ElasticNet-regularized logistic regression on 70 years of data with walk-forward CV, reaching 83% directional accuracy across 50+ MLflow-tracked experiments.
- Served live via FastAPI on Render with a Chart.js frontend showing 7-day predictions, live INR conversion, and confidence-scored trends.

TECHNICAL SKILLS

Languages Python · SQL · R · LaTeX · Mathematica · HTML/CSS/JS

ML / DL PyTorch · TensorFlow/Keras · Scikit-learn · XGBoost · LightGBM

LLMs / Agents LangGraph · LangChain · Gemini & Groq APIs · Transformer Fine-tuning · Prompt Engineering

XAI SHAP · LIME · Activation Patching · Direct Logit Attribution · Bias Detection & Mitigation

MLOps MLflow · DagsHub · DVC · GitHub Actions CI/CD · Docker · Apache Airflow

Deployment FastAPI · MySQL (Aiven) · MongoDB · Streamlit · Cloud (AWS · OCI · Render)

CERTIFICATIONS

Claude Code in Action · Anthropic · Mar 2026 · [Verify ↗](#)

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional · Oracle University · Oct 2025

Oracle Cloud Infrastructure 2025 Certified Data Science Professional · Oracle University · Sep 2025